# Datasets Used in Knowledge Graph & LLM Research

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## Dataset 1: HotpotQA

**Definition**

HotpotQA is a multi-hop question answering dataset where answering a question requires combining information from two or more Wikipedia articles.

**Purpose**

Evaluate:

- Multi-hop reasoning
- Evidence retrieval
- Explainable QA

**Question Types**

- Bridge Questions
- Comparison Questions

**Example**

Question: Where was the author of Harry Potter born?

Harry Potter → J.K. Rowling → Birthplace

**Used In**

- GraphRAG
- HybridRAG
- Think-on-Graph
- Multi-hop LLM evaluation

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## Dataset 2: 2WikiMultiHopQA

**Definition**

A Wikipedia-based multi-hop QA dataset designed to reduce shortcut reasoning by requiring explicit reasoning across multiple articles.

**Purpose**

Evaluate:

- Multi-document retrieval
- Multi-hop reasoning
- Entity linking

**Source:** Wikipedia

**Example**

Tesla → Birth country → Official language

**Used In**

- GraphRAG

- KGQA
- Think-on-Graph

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## Dataset 3: ComplexWebQuestions (CWQ)

**Definition**

A complex QA dataset over the Freebase Knowledge Graph, requiring logical reasoning rather than simple fact lookup.

**Source:** Freebase

**Reasoning**

- Composition
- Conjunction
- Comparison
- Temporal reasoning
- Superlatives

**Example**

Which actors born before 1970 acted in Marvel movies?

**Used In**

- KGQA
- Think-on-Graph
- GraphRAG

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## Dataset 4: WebQuestionsSP (WebQSP)

**Definition**

An extension of WebQuestions that includes semantic parses (logical forms) for each question.

**Purpose**

Evaluate:

- Evaluate semantic parsing and executable logical forms.

**Source:** Freebase

**Example**

Question: Who directed Titanic?

Titanic → Director → James Cameron

**Used In**

- KGQA
- SPARQL generation
- Semantic Parsing
- KELP

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## Dataset 5: GrailQA

**Definition**

A benchmark that evaluates generalizable Knowledge Graph Question Answering, focusing on unseen entities and complex logical reasoning.

**Source:** Freebase

**Question Types**

- Counting
- Comparison
- Composition
- Temporal
- Ordinal

**Example**

Which Nobel Prize winners were born in India?

**Used In**

- Think-on-Graph
- GraphRAG
- KGQA

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## Dataset 6: MetaQA

**Definition**

A movie-domain benchmark for evaluating 1-hop, 2-hop, and 3-hop reasoning over a knowledge graph.

**Source:** Movie Knowledge Graph

**Example**

Who directed the movies acted in by Tom Hanks?

**Used In**

- Think-on-Graph
- KGQA
- Graph Neural Networks

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## Dataset 7: WebQuestions (WebQ)

**Definition**

The original benchmark for answering natural language questions using the Freebase knowledge graph. WebQSP is an improved version of this dataset.

**Source:** Freebase

**Example**

Who wrote Hamlet?

**Used In**

- KGQA
- Early semantic parsing research

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## Dataset 8: SQuAD

**Definition**

The Stanford Question Answering Dataset is a reading comprehension benchmark where answers are extracted from a single Wikipedia paragraph.

**Purpose**

Evaluate:

- Reading comprehension

**Example**

Who discovered penicillin?

**Used In**

- BERT
- RoBERTa
- LLM fine-tuning

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## Dataset 9: Natural Questions (NQ)

**Definition**

Questions collected from real Google search queries with answers found in Wikipedia.

**Example**

Who invented Bluetooth?

**Used In**

- Retrieval-Augmented Generation
- LLM retrieval evaluation

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## Dataset 10: TriviaQA

**Definition**

A large-scale QA dataset collected from trivia websites.

**Example**

Which country has the largest rainforest?

**Used In**

- Open-domain QA
- Retrieval systems

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## Dataset 11: WikiMovies

**Definition**

Movie-domain QA dataset used before MetaQA.

**Purpose**

Evaluate:

- Movie reasoning

**Source:** Movie Knowledge Base

**Used In**

- Memory Networks
- KGQA

## Dataset 12: Freebase

**Definition**

A large collaborative Knowledge Graph created by Metaweb and later acquired by Google. It stores entities and relationships across many domains.

**Type:** Knowledge Graph (not a QA dataset)

**Used In**

- CWQ
- WebQSP
- GrailQA
- WebQuestions
- Think-on-Graph
- KELP

## Dataset 13: Wikidata

**Definition**

A community-maintained structured knowledge graph developed by the Wikimedia Foundation.

**Type:** Knowledge Graph

**Used In**

- GraphRAG
- HybridRAG
- Think-on-Graph
- Modern KGQA

## Dataset 14: DBpedia

**Definition**

A structured knowledge graph automatically extracted from Wikipedia.

**Purpose**

Evaluate:

- Knowledge representation

**Used In**

- KGQA
- Entity Linking
- Graph Neural Networks

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## Dataset 15: YAGO

**Definition**

A large semantic knowledge graph integrating data from Wikipedia, WordNet, and GeoNames.

**Purpose**

Evaluate:

- Entity reasoning

**Used In**

- KGQA
- Think-on-Graph
- KELP

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## Dataset 16: Movie Knowledge Graph (MovieKB)

**Definition**

The knowledge graph used by MetaQA, containing movies, actors, directors, genres, and relationships.

**Used In**

- MetaQA
- Multi-hop reasoning
- Graph Neural Networks

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## Dataset 17: Wikipedia Corpus

**Definition**

The full collection of Wikipedia articles used as an unstructured text corpus rather than a knowledge graph.

**Used In**

- HotpotQA
- 2WikiMultiHopQA
- RAG

- HybridRAG
- GraphRAG

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## Dataset 18: Knowledge Graph Triples

**Definition**

Collections of facts stored as triples: (Head Entity, Relation, Tail Entity). These are not a standalone benchmark but form the basis of many KG systems.

**Example**

(Tesla, bornIn, Smiljan)

**Used In**

- KELP
- Think-on-Graph
- KG-Adapter
- GraphRAG

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## Dataset 19: Entity Linking Benchmarks

**Definition**

Common benchmarks used to evaluate entity linking include AIDA-CoNLL, GERBIL, and TAC-KBP.

**Purpose**

Evaluate:

- Evaluate mapping text mentions to knowledge graph entities.

**Used In**

- GraphRAG
- Think-on-Graph
- HybridRAG

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## Dataset 20: BEIR

**Definition**

The Benchmark for Information Retrieval is a collection of diverse retrieval datasets used to evaluate dense, sparse, and hybrid retrievers.

**Used In**

- HybridRAG
- Dense Retrieval
- BM25 evaluation
- Retriever benchmarking

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## Which Topics Use Which Datasets?

- QA benchmark datasets (e.g., HotpotQA, GrailQA, MetaQA)
- Knowledge graphs (e.g., Freebase, Wikidata, DBpedia, YAGO)
- Text corpora (e.g., Wikipedia)
- Retrieval benchmarks (e.g., BEIR)
- Entity linking benchmarks (e.g., AIDA-CoNLL)